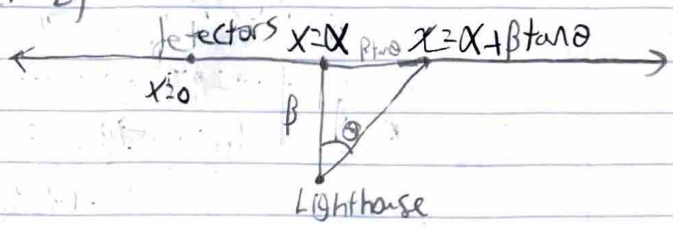
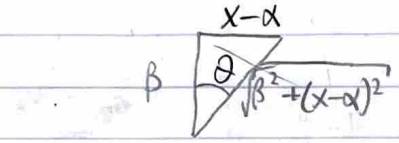
$X = \alpha + \beta \tan\theta$

1. a) ② ~ uniform $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



$X = \alpha + \beta \tan\theta$

$\beta \tan\theta = x - \alpha$.



$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{x-\alpha}{\beta} \Rightarrow \theta = \tan^{-1}\left(\frac{x-\alpha}{\beta}\right)$ see uniform $a=-\frac{\pi}{2}$ $b=\frac{\pi}{2}$

$f(x,\beta) = \frac{d\theta}{dx} = \frac{1}{1+(\frac{x-\alpha}{\beta})^2}\cdot\frac{1}{\beta} = \frac{\beta^2}{\beta^2+(x-\alpha)^2}\cdot\frac{1}{\beta} = \frac{\beta}{\beta^2+(x-\alpha)^2}$

$f(x) = \begin{cases} \frac{1}{b-a} & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$

$\Rightarrow f(\theta) = \begin{cases} \frac{1}{\pi} & -\frac{\pi}{2} \le \theta \le \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$

② independent of α, β since position of lighthouse is not dependent on beam angle and vice versa

$\Rightarrow f(X=x) = f(\theta)\,f(\alpha,\beta) = \frac{\beta}{\pi(\beta^2+(x-\alpha)^2)}$

$X = X_1 + X_2 + \dots X_n$ $\quad X_1, X_2, \dots, X_n$ are iid

Since $X_1, X_2, \dots, X_n$ are iid $\Rightarrow P(X=x) = f(x) = f(x_1)f(x_2)\dots f(x_n)$

$\Rightarrow f(x) = \frac{1}{\pi}\frac{\beta}{\beta^2+(x-\alpha)^2} = \frac{1}{\pi}\frac{\beta}{\beta^2+(x_1-\alpha)^2}\cdot\frac{1}{\pi}\frac{\beta}{\beta^2+(x_2-\alpha)^2}\cdots\frac{1}{\pi}\frac{\beta}{\beta^2+(x_n-\alpha)^2}$

w/ $f(x) \propto \frac{1}{\beta}$ $\quad f(x) \propto \frac{1}{\beta}\prod_{i=1}^{n}\frac{\beta}{\pi(\beta^2+(x_i-\alpha)^2)} = \frac{\beta^{n-1}}{\pi^n}\prod_{i=1}^{n}\frac{1}{\beta^2+(x_i-\alpha)^2}$

Bayes theorem $P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$

$\Rightarrow P(X|\alpha,\beta) = \frac{P(\alpha,\beta|X)\cdot P(X)}{P(\alpha,\beta)} \Rightarrow P(\alpha,\beta|X) = \frac{P(X|\alpha,\beta)\cdot P(\alpha,\beta)}{P(X)}$



$$q(z,y) = \frac{1}{\sqrt{2\pi}} e^{\frac{|z-y|^2}{s^2}}$$

$z_1 = (\alpha_2, b_1)$

$z_2 = (\alpha_1, b_2)$ $\alpha_2 = \alpha_1 +$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

2) $f(x,y) = \frac{1}{\sigma_x\sigma_y\sqrt{2\pi}} e^{-\frac{(x-\mu_x)^2(y-\mu_y)^2}{4\sigma_x^2\sigma_y^2}}$

standard $\sigma_x = \sigma_y = 1$ $\mu_x = \mu_y = 0$

$$f(x,y) = \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{x^2+y^2}{2}\right)}$$

$$f(x) \propto \beta^{n-1} \prod_{i=1}^{n} \frac{1}{\beta^2 + (x_i - \alpha)^2}$$

b) $b = \log(\beta) \Rightarrow \beta = e^b$

$$Z = (\alpha, b) \quad (e^b)^{n-1} \prod_{i=1}^{n} \frac{1}{(e^b)^2 + (x_i-\alpha)^2} = e^{b(n-1)} \prod_{i=1}^{n} \frac{1}{e^{2b} + (x_i-\alpha)^2}$$

$$g(z) = f(\alpha, b) \cdot \frac{d\beta}{db} = \left(e^{b(n-1)} \prod_{i=1}^{n} \frac{1}{e^{2b} + (x_i-\alpha)^2}\right) e^b = e^{bn} \prod_{i=1}^{n} \frac{1}{e^{2b} + (x_i-\alpha)^2}$$

$$S = \frac{1}{\sqrt{2\pi}} \left(e^{-\delta_2^2/2}, e^{-h^2/2}\right)$$

$$g(y) = g(z+s) = g\left(\alpha + \frac{s}{\sqrt{2\pi}} e^{-\frac{\delta^2}{2}}, b + \frac{s}{\sqrt{2\pi}} e^{-h^2/2}\right)$$

$$g(y) = e^{n(b+s\delta_2)} \prod_{i=1}^{n} \frac{1}{e^{2(b+s\delta_2)} + (x_i - (\alpha + s\delta_1))^2}$$

$$\frac{g(y)}{g(z)} = \frac{e^{n(b+s\delta_2)} \prod_{i=1}^{n} \frac{1}{e^{2(b+s\delta_2)} + (x_i - (\alpha+s\delta_1))^2}}{e^{bn} \prod_{i=1}^{n} \frac{1}{e^{2b} + (x_i-\alpha)^2}}$$

$$= e^{ns\delta_2} \prod_{i=1}^{n} \frac{e^{2b} + (x_i-\alpha)^2}{e^{2(b+s\delta_2)} + (x_i - (\alpha+s\delta_1))^2}$$

$\min\left(1, \frac{g(y)}{g(z)}\right)$ $\quad q(z,y) = \frac{1}{\sqrt{2\pi}} e^{\frac{|z-y|^2}{s^2}}$

$g(y)\, q(y,z) = g(z)\, q(z,y)$

$$\frac{g(y)}{g(z)} = e^{ns\delta_2} \prod_{i=1}^{n} \frac{e^{2b} + (x_i-\alpha)^2}{e^{2(b+s\delta_2)} + (x_i - (\alpha+s\delta_1))^2} = \frac{\frac{1}{\sqrt{2\pi}} e^{\frac{|y-z|^2}{s^2}}}{\frac{1}{\sqrt{2\pi}} e^{\frac{|z-y|^2}{s^2}}} = e^{\frac{|y-z|^2}{s^2} - \frac{|z-y|^2}{s^2}} = e^0 = 1$$

$\Rightarrow \frac{g(y)}{g(z)} = 1 \Rightarrow g(y) = g(z)$ $\therefore$ detailed balance is satisfied

Symmetry of the jump is used in $|y-z|^2 = |z-y|^2$

If would sample $f(\alpha, b)$,

c) $\delta = .01$ does not work because it is too small of a step size and does not sample enough of α, β

$\delta = 25$ is too large and skips too much of sample space

2. $\lambda = (3, 2, 1.5, 1.2, 1, .75)$ finish/hr

$M = \max_i X_i$ $\quad X_i \sim \text{exponential}(\lambda_i)$

$\Lambda = \lambda_1 + \dots + \lambda_n$

a) $X = \min_i X_i$, $I = i$ minimizes Goal $P(I=i, X > x) = \frac{\lambda_i}{\Lambda} = \frac{\lambda_i}{\Lambda} e^{-\Lambda x}$

if $X = \min_i X_i$
$\lambda_i < \lambda_{\neq i}$

I is independent of X as property of Poisson Property $\Rightarrow P(I=i, X>x) = P(I=i)P(X>x)$

$P(I=i) = \frac{\lambda_i}{\Lambda}$ $\frac{\text{(size of RV)}}{\text{(size of sample space)}}$

$P(X > x) = P(X_i > x_i)$

$F(x) = P(\min(X_1, X_2 \dots, X_n) \le x)) = 1 - P(\min(X_1, X_2 \dots, X_n) \ge x)$

$X_1, X_2, \dots X_n$ are ind.

$\Rightarrow 1 - P(X_1 \ge x)P(X_2 \ge x) \cdots P(X_n \ge x) = 1 - e^{-\lambda_1} e^{-\lambda_2} e^{-\lambda_3} \cdots e^{-\lambda_n}$

$= 1 - e^{-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)} = 1 - e^{-\Lambda} = \text{Xi exp}(\Lambda)$

let $i = 1$

$P(X = x_1) = P(\min = x_1, x_2 \dots x_n) = P(X_1 \ge x_2, \dots x_n) = \int_0^{\infty} \int_0^{x_{n-1}} \cdots \int_0^{x_2} f(x_1, x_2, \dots x_n) \, dx_1 \cdots dx_n$

$= \frac{\lambda_1}{\lambda_1 + \lambda_2 + \cdots \lambda_n} = \frac{\lambda_1}{\Lambda}$ generalized to $\frac{\lambda_i}{\Lambda} = P(I=i)$

$P(I=i, x>x) = P(I=i)P(x>x) = \frac{\lambda_i}{\Lambda} e^{-\Lambda x}$

b) memoryless: $P(X>X_1+X_2) = P(X>X_2)$ w/ $X_2>X_1$

$P(X_{\sigma_n} > X_{\sigma_{n-1}} \cdots > X_{\sigma_1}) = P(X_{\sigma_n} > X_{\sigma_{n-1}}) \cdot P(X_{\sigma_{n-1}} > X_{\sigma_{n-2}}) \cdots P(X_{\sigma_2} > X_{\sigma_1})$

since X_i are independent

$$\int_0^{\infty}\int_0^{x_{\sigma_{n-1}}} \lambda_{\sigma_n}\lambda_{\sigma_{n-1}} e^{-(\lambda_{\sigma_n}x_{\sigma_n} + \lambda_{\sigma_{n-1}}x_{\sigma_{n-1}})} \, dx_{\sigma_n}\, dx_{\sigma_{n-1}} \cdots$$

$$= \frac{\lambda_{\sigma_n}}{\lambda_{\sigma_n}+\lambda_{\sigma_{n-1}}} \cdot \frac{\lambda_{\sigma_{n-1}}}{\lambda_{\sigma_{n-1}}+\lambda_{\sigma_{n-2}}} \cdots \frac{\lambda_{\sigma_2}}{\lambda_{\sigma_2}+\lambda_{\sigma_1}}$$

$\lambda = (1,2,3)$ $\quad \Lambda = 1+2+3 = 6$

$$\prod_{k=1}^{3} = \frac{3}{6} \cdot \frac{2}{6-1} \cdot \frac{1}{6-1-2}$$

$$= \frac{1}{2} \cdot \frac{2}{5} \cdot \frac{1}{3} = \boxed{\frac{2}{30}}$$

c) $P(M \le t) = P(\max(X_i) \le t) = P(X_1 \le t, X_2 \le t, \ldots, X_n \le t)$

$M = \max(X_i)$ since X_i independent

$$\Rightarrow = P(X_1 \le t)\,P(X_2 \le t) \cdots P(X_n \le t)$$
$$= (1-P(X_1 \ge t))(1-P(X_2 \ge t)) \cdots (1-P(X_n \ge t))$$
$$= (1-e^{-\lambda_1 t})(1-e^{-\lambda_2 t}) \cdots (1-e^{-\lambda_n t})$$

$$\boxed{P(M \le t) = \prod_i (1-e^{-\lambda_i t})}$$

for $i = 1:3$

$$\prod_{i=1}^{3}(1-e^{-\lambda_i t}) = (1-e^{-\lambda_1 t})(1-e^{-\lambda_2 t})(1-e^{-\lambda_3 t}) = (1 - e^{-\lambda_2 t} - e^{-\lambda_1 t} + e^{-(\lambda_1+\lambda_2)t})(1-e^{-\lambda_3 t})$$

$$P(M>t) = 1 - e^{-\lambda_3 t} - e^{-\lambda_2 t} - e^{-\lambda_1 t} + e^{-(\lambda_1+\lambda_2)t} - e^{-(\lambda_1+\lambda_2+\lambda_3)t} + e^{-(\lambda_2+\lambda_3)t} + e^{-(\lambda_1+\lambda_3)t}$$

$$\int_{-\infty}^{\infty} t\,P(M>t)\,dt = \lim_{\alpha\to\infty}\int_{-\alpha}^{\alpha} t\,P(M>t)\,dt = 2\lim_{\alpha\to\infty}\left[\frac{t^2}{2}\right]_0^{\alpha} + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3} - \frac{1}{\lambda_1+\lambda_2} - \frac{1}{\lambda_1+\lambda_3} - \frac{1}{\lambda_2+\lambda_3} + \frac{1}{\lambda_1+\lambda_2+\lambda_3}$$

$$\boxed{E[M] = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3} - \frac{1}{\lambda_1+\lambda_2} - \frac{1}{\lambda_1+\lambda_3} - \frac{1}{\lambda_2+\lambda_3} + \frac{1}{\lambda_1+\lambda_2+\lambda_3}}$$

$$\prod_{k=1}^{2} \frac{\lambda_{\sigma_k}}{\Lambda - \lambda_{\sigma_1} - \cdots \lambda_{\sigma_{k-1}}} = \frac{\lambda_{\sigma_1}}{(\lambda_1+\lambda_2)} \cdot \frac{\lambda_{\sigma_2}}{\lambda_1+\lambda_2-\lambda_{\sigma_1}}$$

since this yields $\frac{\lambda_{\sigma_2}}{\lambda_{\sigma_2}} = 1$

$$E[N] = \frac{\lambda_2(\lambda_1+\lambda_2)}{\lambda_2\lambda_1(\lambda_1+\lambda_2)} + \frac{\lambda_1(\lambda_1+\lambda_2)}{\lambda_1\lambda_2(\lambda_1+\lambda_2)} - \frac{\lambda_1\lambda_2}{\lambda_1\lambda_2(\lambda_1+\lambda_2)} = \frac{2\lambda_1\cdot\lambda_2 + \lambda_1^2+\lambda_2^2-\lambda_1\lambda_2}{\lambda_1\lambda_2(\lambda_1+\lambda_2)} = \frac{\lambda_1^2+\lambda_2^2-\lambda_1\lambda_2}{\lambda_1\lambda_2(\lambda_1+\lambda_2)}$$

3 a) $\lambda = 4$/hour hppp $t_1, t_2 \ldots$ are Exp(λ)

$T_n = t_1 + t_2 + \ldots + t_n$ $N(t) = \max\{n : T_n \le t\}$

$$\frac{d}{dt}\left[1 - \sum_{k=0}^{n-1} e^{-\lambda t}\frac{(\lambda t)^k}{k!}\right] = \frac{d}{dt}(1) - \frac{d}{dt}\sum_{k=0}^{n-1} e^{-\lambda t}\frac{(\lambda t)^k}{k!}$$

$$= -\sum_{k=0}^{n-1} \frac{d}{dt}\left(e^{-\lambda t}\frac{(\lambda t)^k}{k!}\right) = -\sum_{k=0}^{n-1}\left(-\frac{1}{\lambda}e^{-\lambda t}\left(\frac{(\lambda t)^k}{k!}\right) + \frac{k e^{-\lambda t}\lambda^k t^{k-1}}{k!}\right)$$

Product rule

$$= \sum_{k=0}^{n-1} \frac{\lambda^k e^{-\lambda t} t^{k-1}}{(k-1)!} - \frac{\lambda^{k-1} e^{-\lambda t} t^k}{k!} = e^{-\lambda t}\sum_{k=0}^{n-1}\frac{\lambda^k t^{k-1}}{(k-1)!} - \frac{\lambda^{k-1} t^k}{k!}$$

$$= e^{-\lambda t}\sum_{k=0}^{n-1}\frac{(\lambda t)^{k-1}}{(k-1)!}\left(\lambda - \frac{t}{k}\right) = \lambda e^{-\lambda t}\sum_{k=0}^{n-1}\left(1 - \frac{\lambda t}{k}\right)\frac{(\lambda t)^{k-1}}{(k-1)!}$$

$$= \lambda e^{-\lambda t}\underbrace{\sum_{k=0}^{n-1}\frac{(\lambda t)^{k-1}}{(k-1)!} - \frac{(\lambda t)^k}{(k-1)!}}_{\text{telescoping series}} \to \frac{(\lambda t)^{n-1}}{(n-1)!}$$

$$= \boxed{\lambda e^{-\lambda t}\frac{(\lambda t)^{n-1}}{(n-1)!}}$$

$\lambda = 4/\text{hour} = \frac{1}{15}$/min

$N(t)$ is counting parameter

$$\boxed{P(N(t)=0) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\frac{1}{15}} = .9355}$$

$P(T_n \le t) = P(N(t) \ge n) = P(N(t) \ge 3) = 1 - P(T_n \ge t)$ for $n=3, t=60$

$$= \boxed{P(N(60)=3) = \frac{(\lambda t)^3 e^{-\lambda}}{3!} = \frac{(\frac{1}{15}\cdot 60)^3 e^{-\frac{1}{15}\cdot 60}}{3!} = .1954}$$

3. b) $N(1)=3 \Rightarrow P(N(1)=3) = \frac{(\lambda t)^3 e^{-\lambda t}}{3!}$ w/ $t=1$ $= \frac{(\lambda)^3 e^{-\lambda}}{3!} = \frac{(2)^3 e^{-\lambda}}{6} = 2$

$N(\frac{1}{2}) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$ @ $t=\frac{1}{2}$ $\frac{\frac{\lambda^n}{2^n} e^{-\frac{\lambda}{2}}}{n!}$

$P(N(\frac{1}{2}) | N(1)=3)$ since independent

$= \frac{\frac{\lambda^n}{2^n} e^{-\frac{\lambda}{2}}}{n!} \cdot \frac{\lambda^3 e^{-\lambda}}{3!} = (\frac{1}{2})^n \frac{\lambda^{n+3} e^{-(\frac{3}{2}\lambda)}}{n!3!} = \lambda^{n+3} e^{-\frac{3}{2}\lambda} \cdot \frac{(\frac{1}{2})^n}{n!3!}$

Binomial

$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

w/ $p=\frac{1}{2}$ $x=3$ $\frac{n!}{3!(n-3)!}$

$= \binom{n}{3} \frac{1}{2}^x \frac{1}{2}^{n-x} = \binom{n}{3} \frac{1}{2}^n$

c)

$f(t_1, t_2) = \lambda e^{-\lambda t_1} \lambda e^{-\lambda(t_2-t_1)} e^{-\lambda(t-t_2)}$

$P(N(t)=2) = \frac{(\lambda t)^2 e^{-\lambda t}}{2!}$

$\frac{f(T_1, T_2)}{P(N(t)=2)} = \frac{\lambda^2 e^{-\lambda(t_2+t_1-t_1+t-t_2)}}{\frac{(\lambda t)^2 e^{-\lambda t}}{2!}} = \frac{2}{t^2} \Rightarrow \frac{n}{t^n}$

Simulate the Process using $\frac{n}{t^n}$

d) CDF exponential $\Rightarrow F(x) = 1 - e^{-\lambda x}$

$F^{-1}(U) = 1 - e^{-\lambda F^{-1}}$

$\Rightarrow 1 - U = e^{-\lambda F^{-1}}$

$\Rightarrow \ln(1-U) = -\lambda F^{-1}$ $\quad U \sim \text{uniform}(0,1)$

$\Rightarrow F^{-1}(U) = \frac{-\ln(1-U)}{\lambda}$

For $N(1)$ $\quad \lambda t = \lambda \Rightarrow$ Poisson(λ)

$\lambda = 4$

$P(N(1) = n) = \frac{\lambda^n e^{-\lambda}}{n!} = \frac{4^n e^{-4}}{n!}$